

Finding Borders

Time limit: 1.00 s **Memory limit:** 512 MB

A *border* of a string is a prefix that is also a suffix of the string but not the whole string. For example, the borders of `abcababcb` are `ab` and `abcb`.

Your task is to find all border lengths of a given string.

Input

The only input line has a string of length n consisting of characters `a–z`.

Output

Print all border lengths of the string in increasing order.

Constraints

- $1 \leq n \leq 10^6$

Example

Input:
`abcababcb`

Output:
`2 5`